

Saad Khan

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PROFESSIONAL SUMMARY

ASIC Design and Verification Engineer with 2+ years of industry experience delivering commercial ASIC and FPGA projects across RTL Design, SystemVerilog/UVM Verification, Design-for-Test (DFT), FPGA prototyping, and open-source processor development. Experienced in architecting reusable verification environments, leading verification teams, achieving coverage closure, and developing synthesizable RTL including RISC-V, IEEE 1149.1 JTAG and IEEE 1687 IJTAG infrastructures. Proficient with Synopsys and Siemens EDA tool flows from RTL design through verification sign-off.

TECHNICAL SKILLS

Languages	SystemVerilog, Verilog, C/C++, Python, Bash
RTL Design	RTL Architecture, FPGA Design, RISC-V, Pipelining, Hazard Detection, Branch Prediction, FSM Design, Parameterized IP Development
Verification	UVM, Constrained Random Verification, Functional Coverage, Code Coverage, Coverage Closure, Scoreboards, Assertions (SVA), X-Propagation, Performance Testing, Time Profiling
Protocols	AXI4, AXI-Stream, APB, UART, AMBA Protocols, CDC Basics
DFT	IEEE 1149.1 JTAG, IEEE 1687 IJTAG, Boundary Scan, TAP Controller, Segment Insertion Bits (SIB), Test Data Registers (TDR), Scan Architecture
EDA Tools	Siemens QuestaSim, Synopsys VCS, Xilinx Vivado, Quartus Prime, Verdi, DVE
Automation	Git, Github Actions, Docker, Linux, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Logic Silicon (formerly DreamBig Semiconductors, acquired by Arm) <i>Associate Hardware Engineer</i>	Islamabad, Pakistan Jun 2026 – Present
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- Leading subsystem verification of the DFT block for Intel Altera's FMMV FPGA program, developing reusable UVM environments, verification plans, coverage models, and regression infrastructure.
- Owned verification planning and sign-off for the DFT subsystem, developing reusable UVM components, verification plans, coverage models, regressions, and debug infrastructure across multiple IP features.
- Achieved verification closure through functional/code coverage analysis, regression automation, and cross-functional debug with RTL and architecture teams.

<i>Junior Hardware Engineer</i>	Dec 2024 – Jun 2026
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- Led a 4-engineer verification team for the Mercury C3PC project, coordinating verification planning, task allocation, sprint execution, and technical reviews for an 8-engineer development effort.
- Architected a reusable UVM verification environment including agents, scoreboards, reference models, coverage collectors, and reusable sequences, supporting multi-million packet simulations across 64 isolation domains with over 8,000 destination IDs.
- Increased functional coverage from 45% to 100% through interface-level covergroups, cross coverage, targeted test development, and regression-driven closure.
- Developed reusable protocol-compliant AXI4, AXI-Stream, and APB UVM agents supporting constrained-random stimulus, protocol checking, and scalable subsystem integration.
- Reduced local regression runtime from 30 minutes to 5 minutes and CI regression runtime from 40 minutes to under 4 minutes by optimizing regression flow, automation scripts, and nightly execution.
- Identified, reproduced, and tracked more than 85 RTL design bugs through closure while collaborating closely with design teams.

NUST Chip Design Center (NCDC) <i>RTL Design Engineer</i>	Islamabad, Pakistan Jun 2024 – Dec 2024
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- Designed synthesizable SystemVerilog RTL and FPGA prototypes for parameterized hardware IP including UARTs, arithmetic units, and custom RISC-V processors.
- Developed single-cycle and pipelined RISC-V processors featuring hazard detection, forwarding, branch prediction, and timing-aware architectures.
- Implemented UART IP with configurable baud-rate generators, FIFOs, READY/VALID interfaces, and FPGA validation.
- Managed multiple RTL development teams while mentoring engineers on FPGA prototyping, RTL coding practices, verification methodology, and clock-domain crossing fundamentals.

CHIP-DV Lab, GIKI <i>Research Assistant</i>	Topi, Pakistan Jun 2023 – Jun 2024
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- Designed and FPGA prototyped single-cycle and 5-stage pipelined RISC-V processors featuring hazard detection, forwarding logic, branch handling, and RV32I support.

- Led the RTL-to-GDSII implementation of two successful TinyTapeout ASIC projects using the Sky130 process, completing synthesis, floorplanning, placement, routing, timing closure, LVS, and DRC sign-off.
- Developed reusable RTL IP including UART controllers, Booth multipliers, Wallace Tree multipliers, and verification environments using SystemVerilog and UVM.
- Delivered technical seminars and laboratory sessions on ASIC Design, FPGA Prototyping, SystemVerilog, UVM, and RISC-V processor architecture for undergraduate engineering students.

Musketeers Tech

Artificial Intelligence Intern

Lahore, Pakistan

Jun 2023 – Sep 2023

- Developed end-to-end AI solutions including an Urdu Text-to-Speech model, multilingual chatbot, and image classification pipeline using PyTorch, TensorFlow, HuggingFace, and OpenAI APIs.

OPEN-SOURCE RTL PROJECTS

IEEE 1149.1 JTAG Controller (SystemVerilog) [GitHub]

2025

- Architected a fully synthesizable, modular IEEE 1149.1 compliant JTAG TAP Controller supporting Boundary Scan Register, Instruction Register, BYPASS, IDCODE, SAMPLE, PRELOAD, EXTEST, INTEST, and CLAMP instructions.
- Partitioned the complete TAP architecture into reusable synthesizable RTL modules including TAP FSM, Instruction Register, Boundary Scan Register, Boundary Scan Cells, TDR decoder, and parameterized shift registers.
- Verified the complete implementation using a commercial JTAG verification environment with functional coverage and regression testing.
- Designed the architecture to support seamless migration toward IEEE 1687 (IJTAG) through modular scan-path partitioning and reusable scan infrastructure.

IEEE 1687 IJTAG Network (SystemVerilog) [GitHub]

2026

- Developed a synthesizable IEEE 1687 (IJTAG) network featuring hierarchical Segment Insertion Bits (SIBs), configurable Test Data Registers (TDRs), and dynamic scan-path reconfiguration.
- Implemented reusable RTL for SIBs, parameterized TDRs, hierarchical instrument networks, and configurable scan segments supporting runtime insertion and bypass.
- Developed a complete SystemVerilog testbench demonstrating hierarchical SIB programming, multi-stage instrument activation, and end-to-end IJTAG scan transactions.

INTELLERA — Hardware Accelerated Matrix MAC Processor [GitHub]

- Designed a custom 5-stage pipelined RISC-V processor implementing RV32I along with 11 custom Matrix-MAC instructions for hardware acceleration.
- Integrated a systolic-array based Matrix MAC engine, custom decoder, forwarding logic, hazard detection, and branch prediction.
- Developed a UART-based instruction loader capable of dynamically loading executable programs without FPGA reconfiguration.

Parameterized Skid Buffer IP [GitHub]

- Developed a configurable SystemVerilog skid buffer supporting variable widths, depths, READY/VALID flow control, back-pressure handling, and reusable protocol-independent interfaces.

Open-Source RISC-V Processors [5-Stage] / [Single-Cycle]

- Developed both single-cycle and fully pipelined RISC-V processors supporting FPGA deployment, hazard detection, forwarding, branch prediction, and reusable RTL architectures.

ASIC TAPEOUTS

8-bit UART Transceiver — TinyTapeout 6 (Sky130)

Directed RTL design, verification, FPGA prototyping, and ASIC implementation of a full-duplex UART supporting configurable baud rates and oversampling. Successfully fabricated through Efabless TinyTapeout 6.

24-Hour Digital Clock — TinyTapeout 6 (Sky130)

Completed RTL-to-GDSII implementation using OpenLane including synthesis, physical design, timing closure, LVS, DRC, and successful fabrication.

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Bachelor of Science — Computer Engineering

2020 – 2024

- Academic High Distinction in the final year (SGPA: 3.81 and 3.79).

CERTIFICATIONS

- Cadence — Design for Test Fundamentals
- Cadence — Essential SystemVerilog for UVM
- Cadence — SystemVerilog Accelerated Verification with UVM
- GIKI — Application Specific Integrated Circuits (ASIC)